Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Problem

Draw a hypocycloid of a circle of diameter 50 mm, which rolls inside a circular arc of radius 85 mm for one revolution. Also, draw a tangent and a normal to the epicycloid at a point 50 mm from the centre of the directing circle.

Calculated Data

$$R = 85 \text{ mm}, \qquad r = \frac{50}{2} = 25 \text{ mm},$$

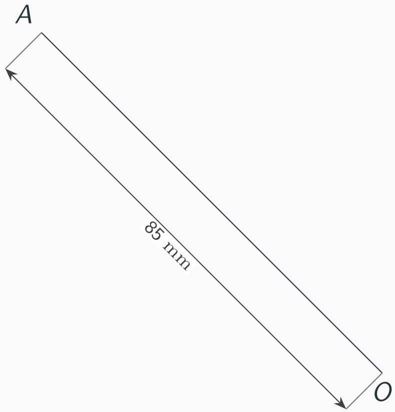
$$OC_0 = R - r = 85 - 25 = 60 \text{ mm}, \qquad AC_0 = 25 \text{ mm},$$

$$\angle AOB = 360^\circ \frac{r}{R} = 360^\circ \frac{25}{85} = 105.882353^\circ,$$

$$\Delta\theta = \frac{105.882353^\circ}{12} = 8.823529^\circ, \qquad \text{Generating-circle division} = 30^\circ.$$

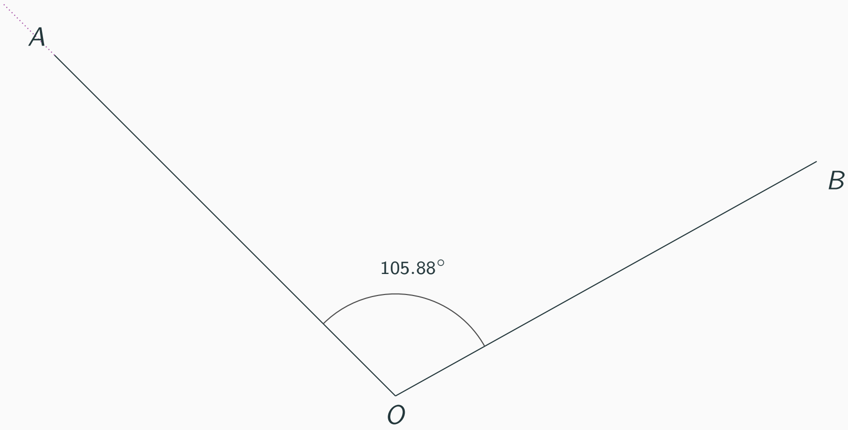
Step 1: Draw OA

$OA = 85$ mm.

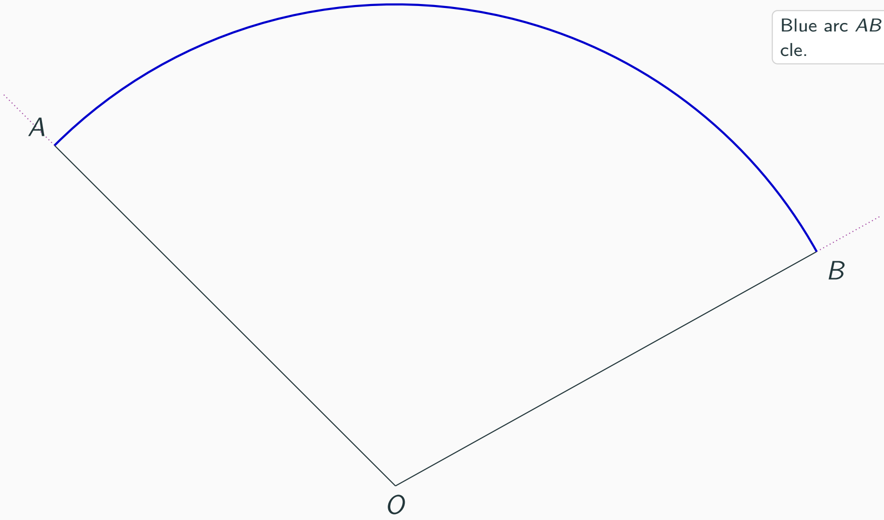


Step 2: Draw *OB* at 105.88°

$\angle AOB = 105.88^\circ$.

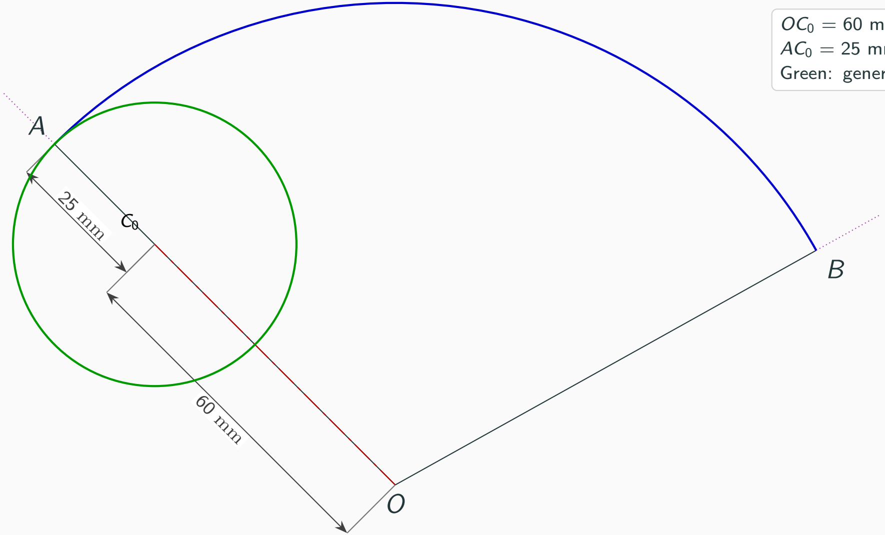


Step 3: Draw the directing arc AB



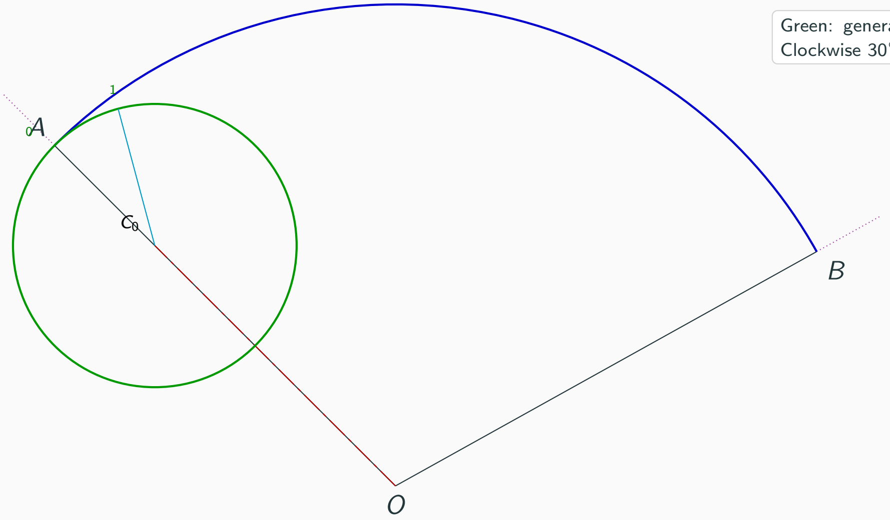
Blue arc AB: directing circle.

Step 4: Locate C_0 and the generating circle



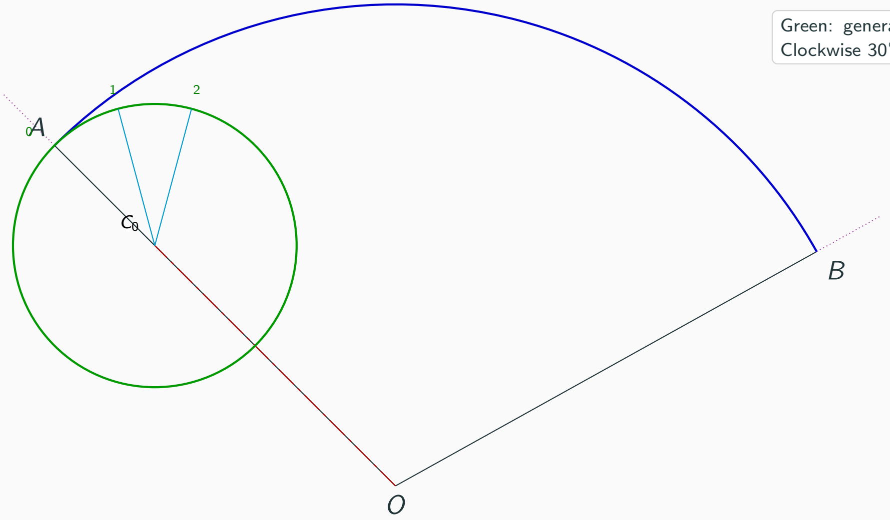
$OC_0 = 60$ mm.
$AC_0 = 25$ mm.
Green: generating circle.

Step 5: Divide the generating circle into 12 equal parts



Green: generating circle.
Clockwise 30° divisions.

Step 5: Divide the generating circle into 12 equal parts



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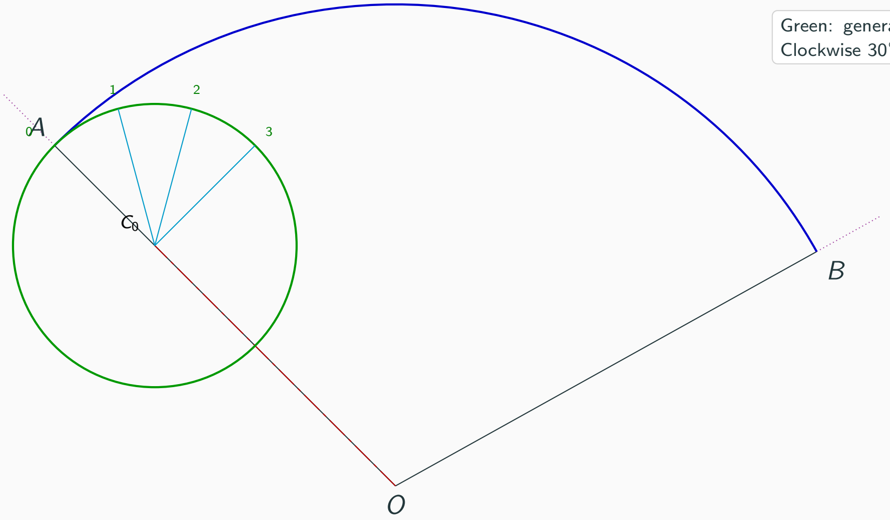

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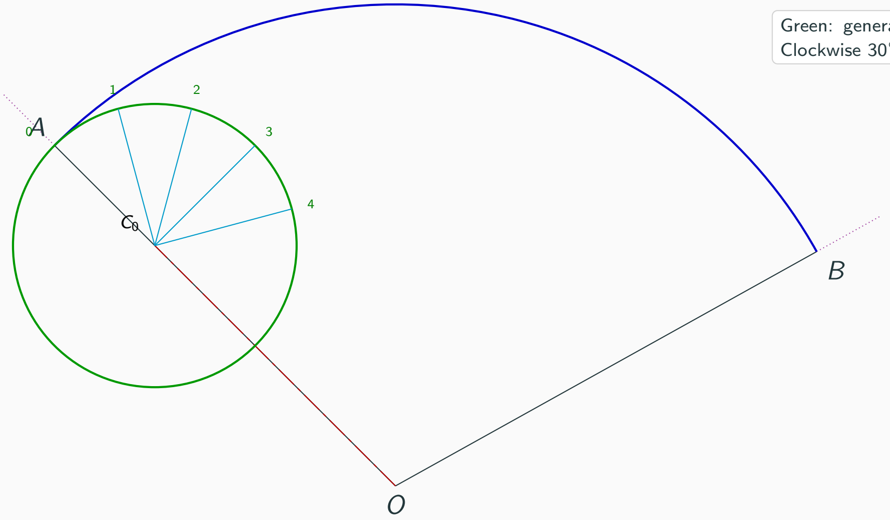

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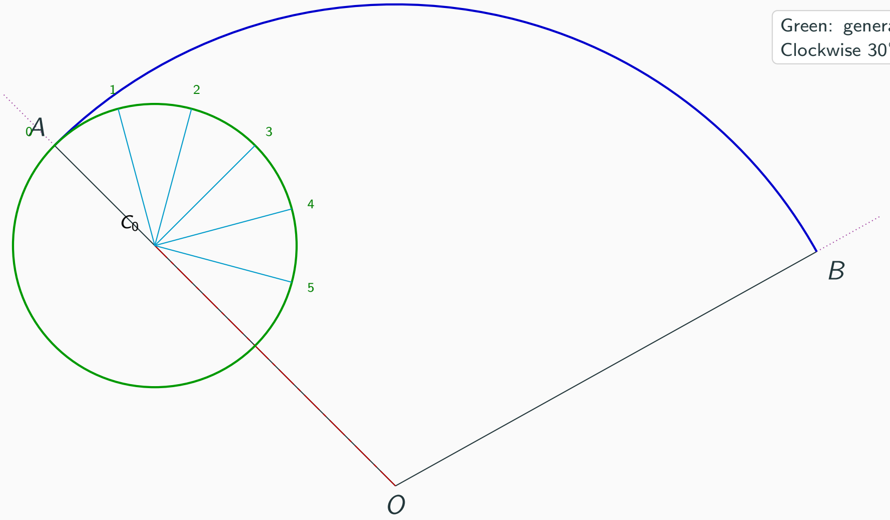

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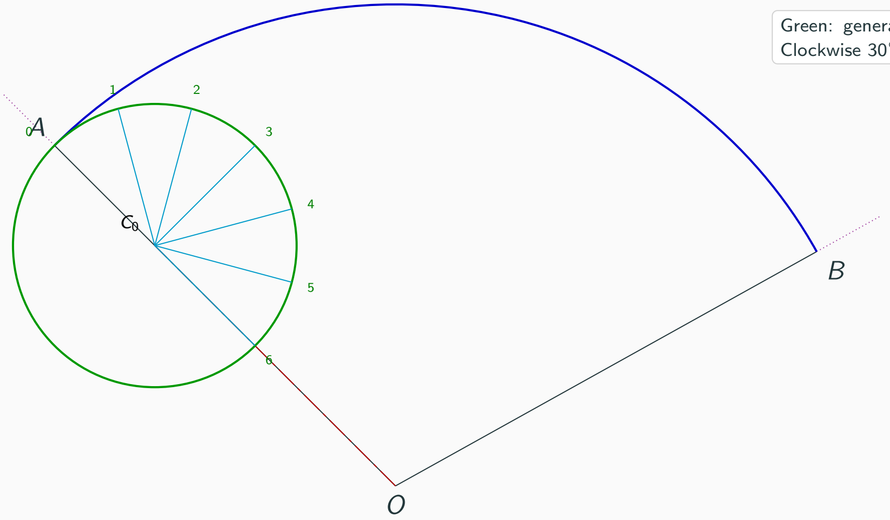

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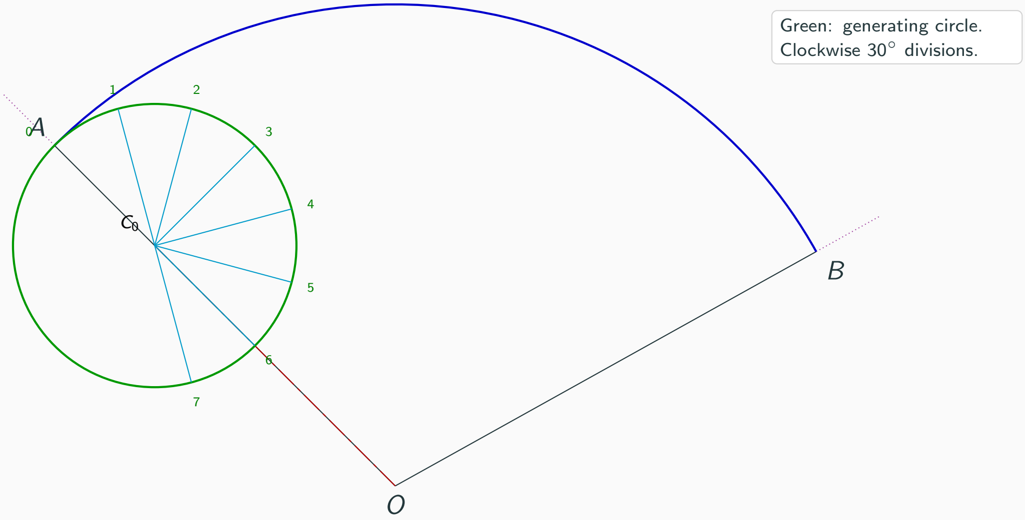

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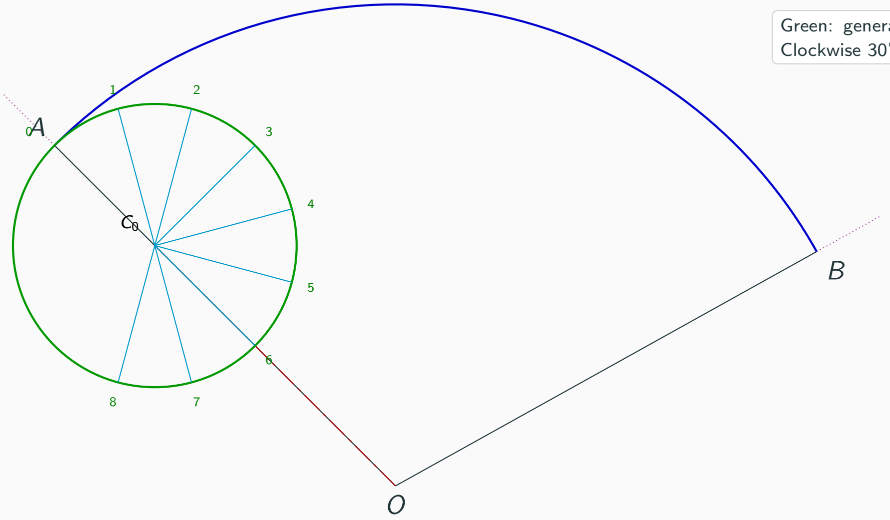

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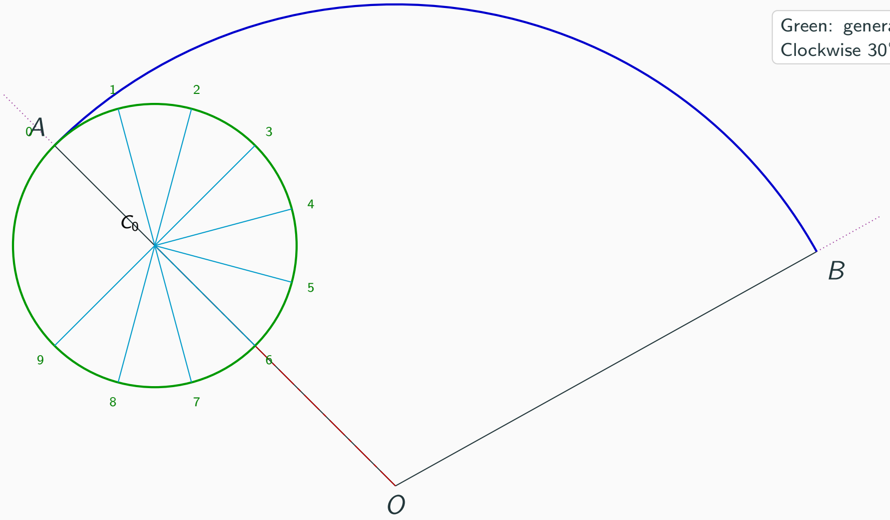

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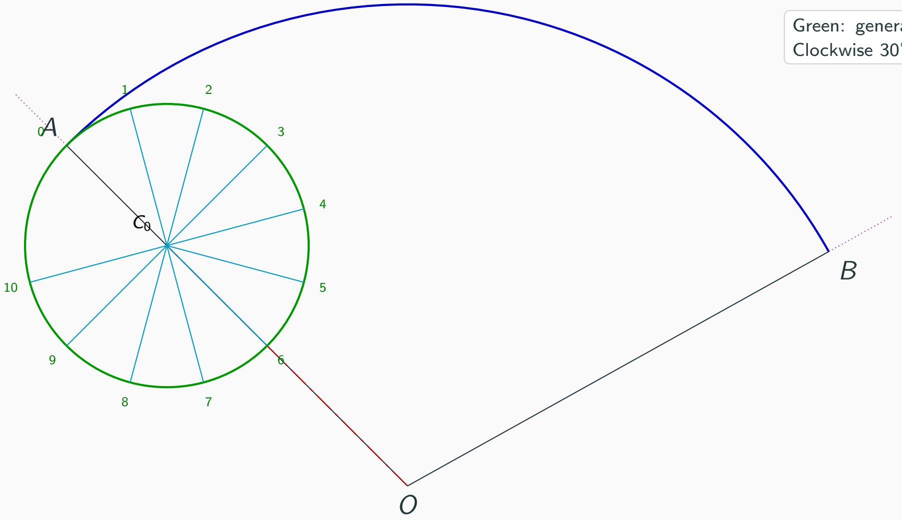

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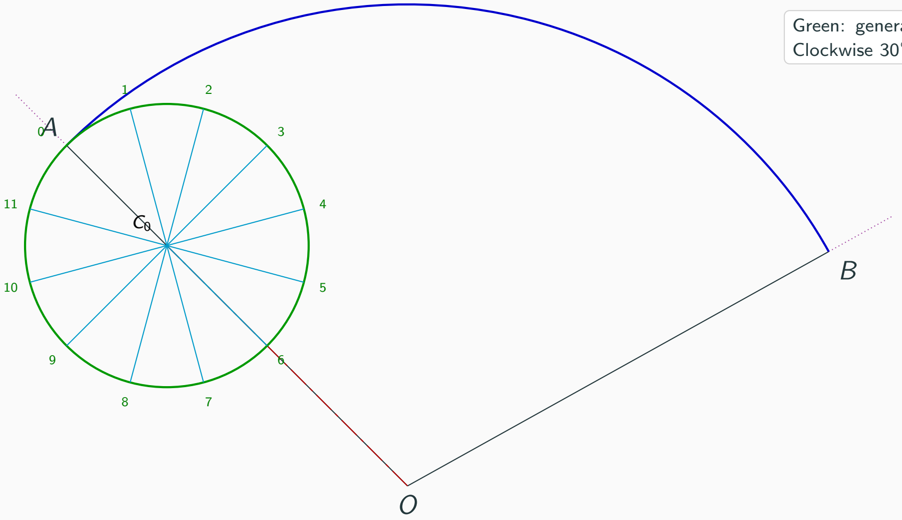

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Clockwise 30° divisions.

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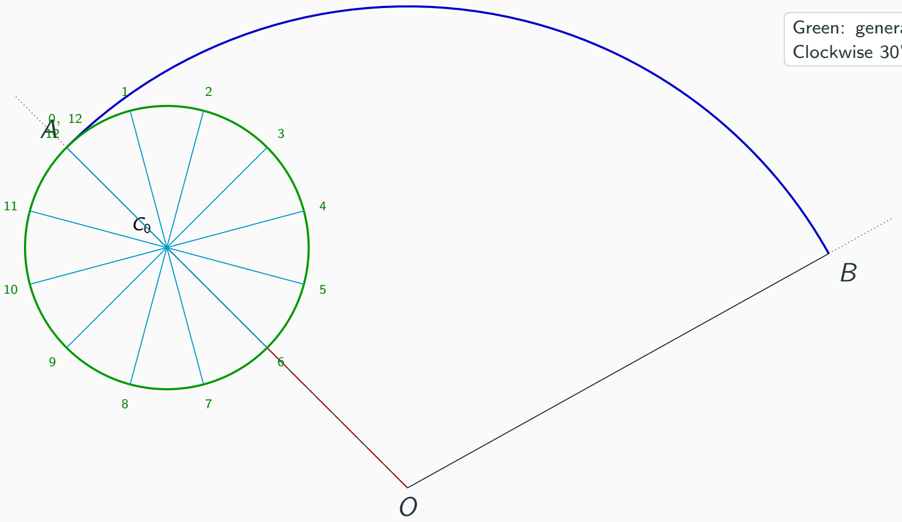
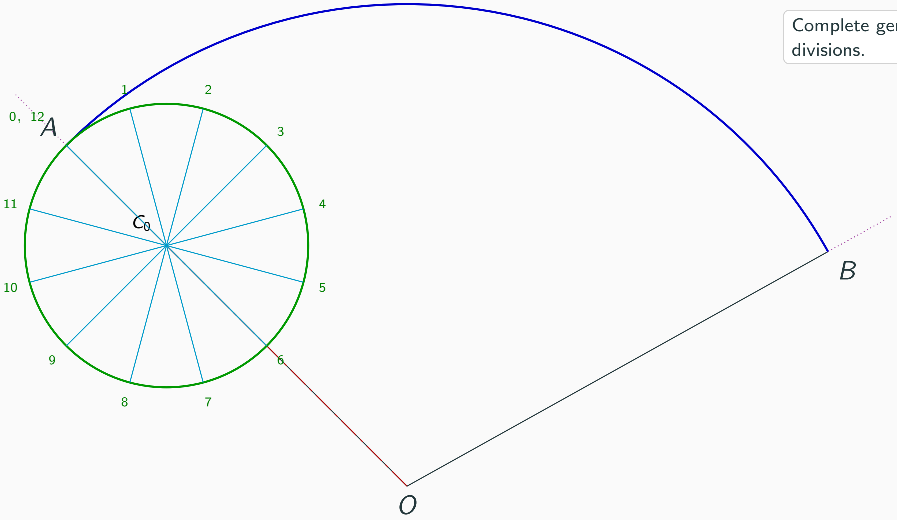


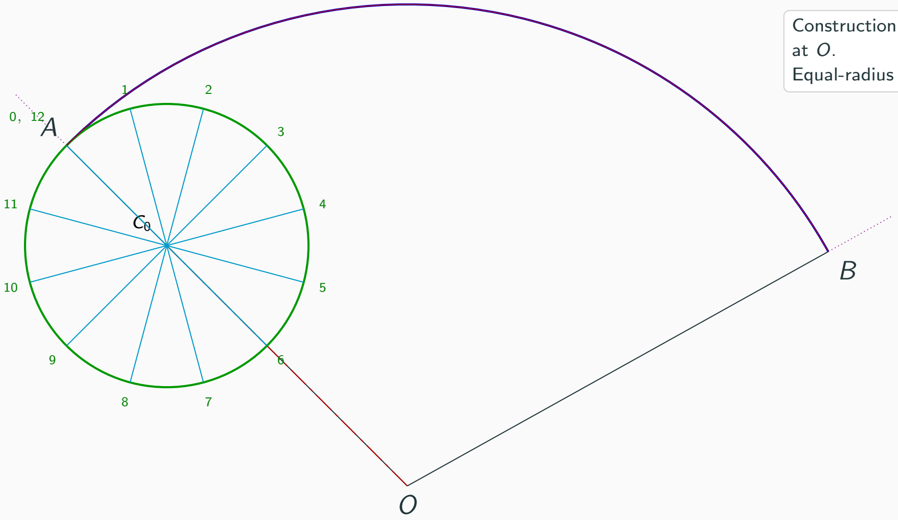
Green: generating circle.
Clockwise 30° divisions.

Step 6: Complete the generating-circle divisions



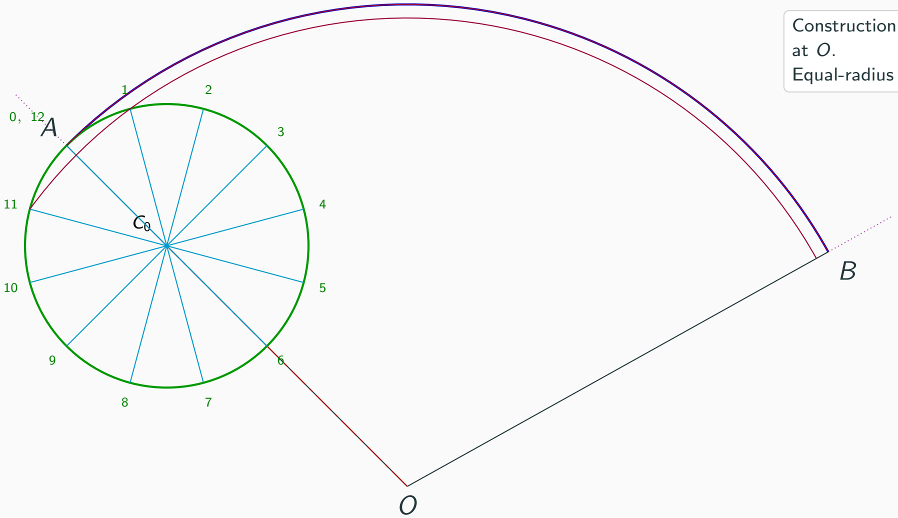
Complete generating-circle divisions.

Step 7: Draw concentric construction arcs



Construction arcs centered at O.
Equal-radius pairs are used.

Step 7: Draw concentric construction arcs



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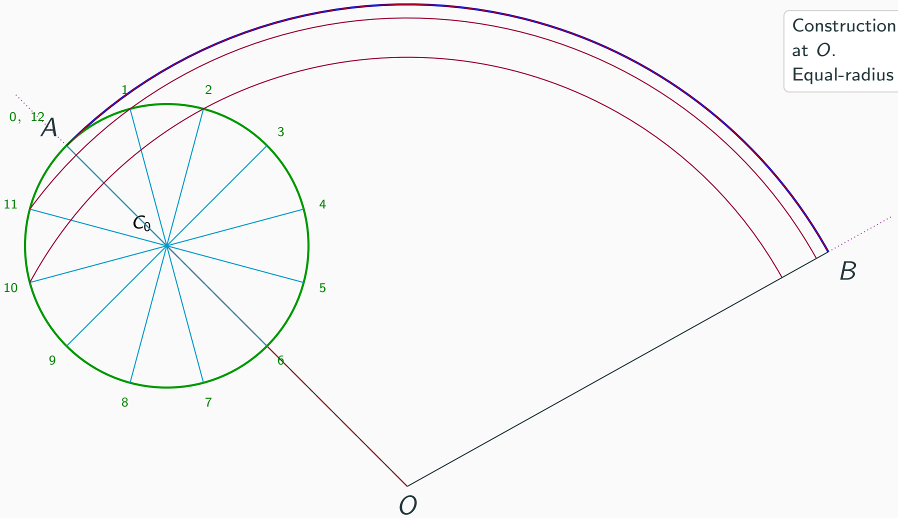

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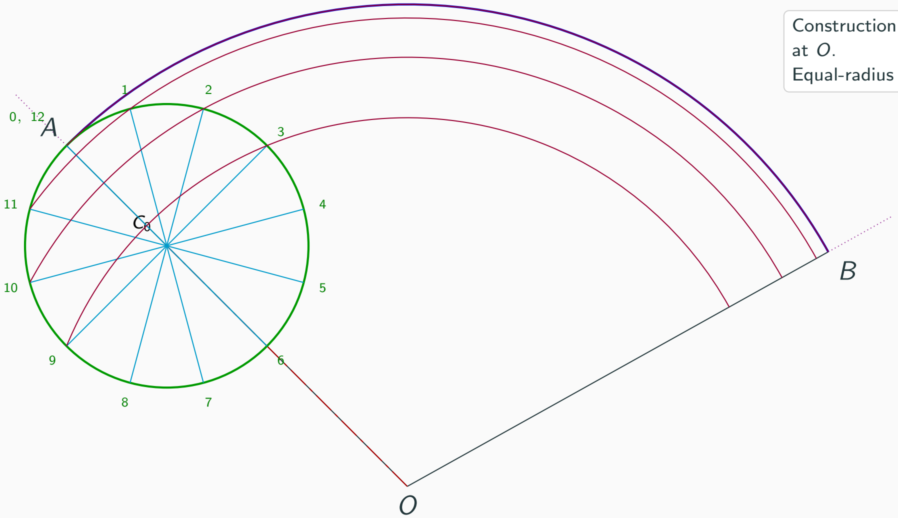

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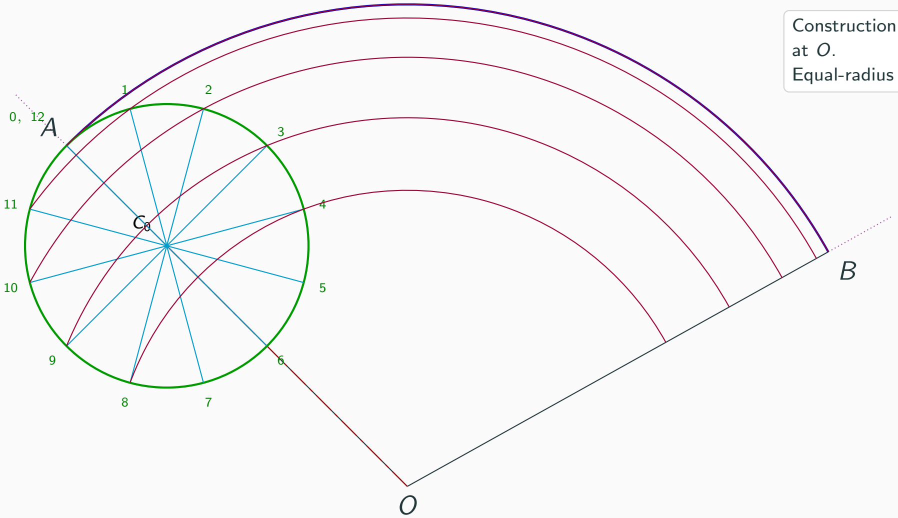

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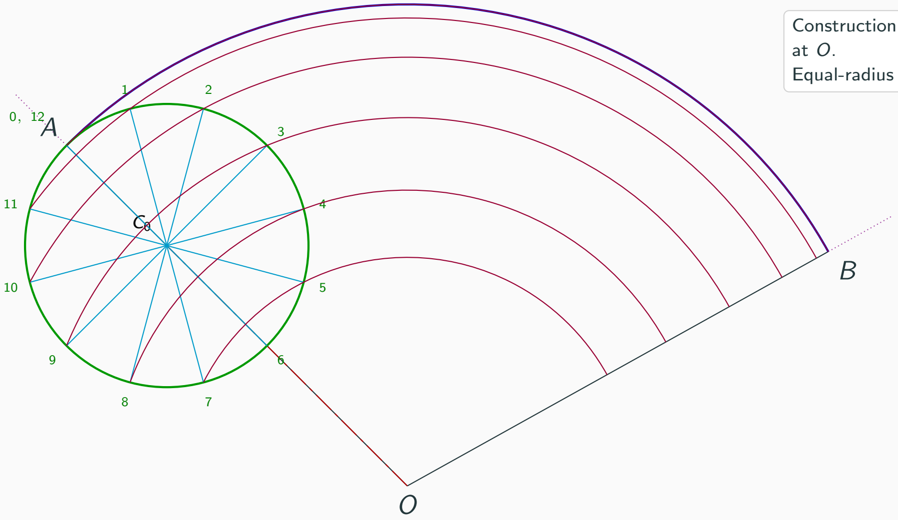

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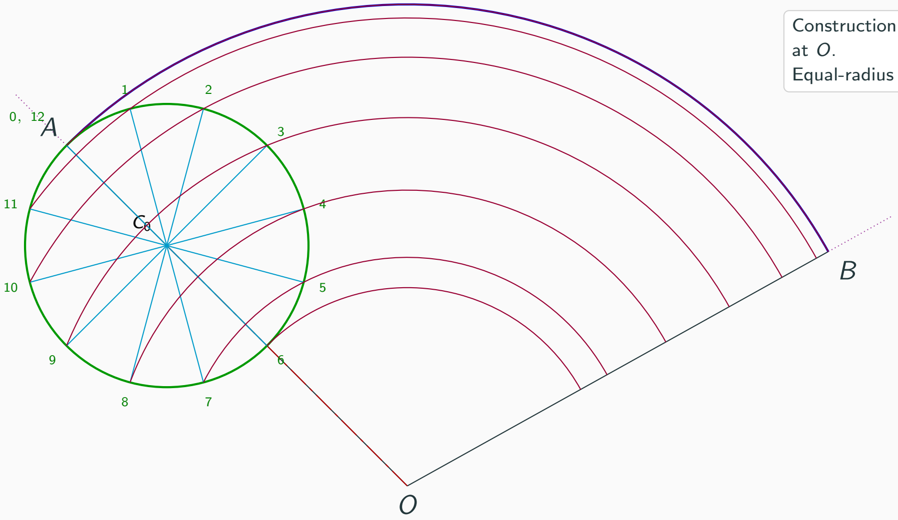
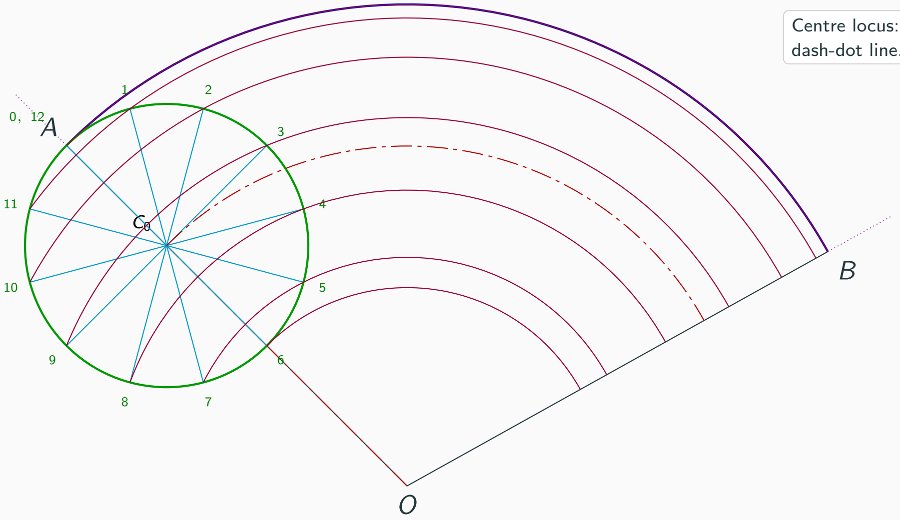


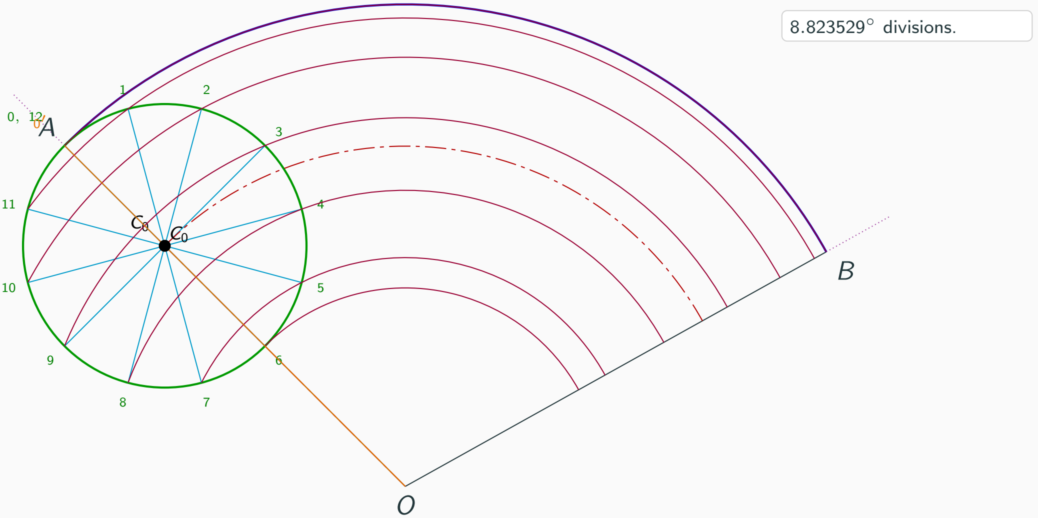
Construction arcs centered at O.
Equal-radius pairs are used.

Step 8: Draw the centre arc through C_0



Centre locus: thin ISO-style dash-dot line.

Step 9: Divide $\angle AOB$ into 12 equal parts



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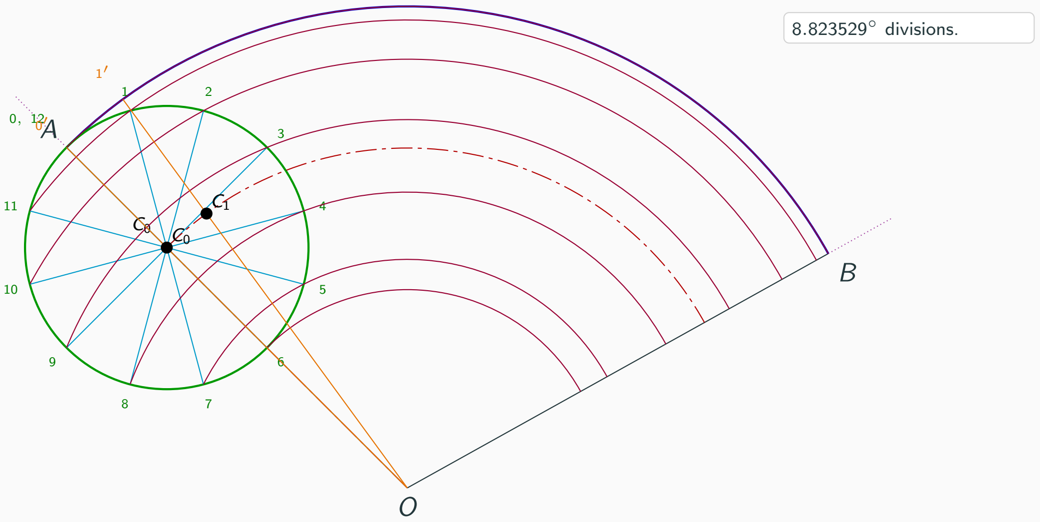

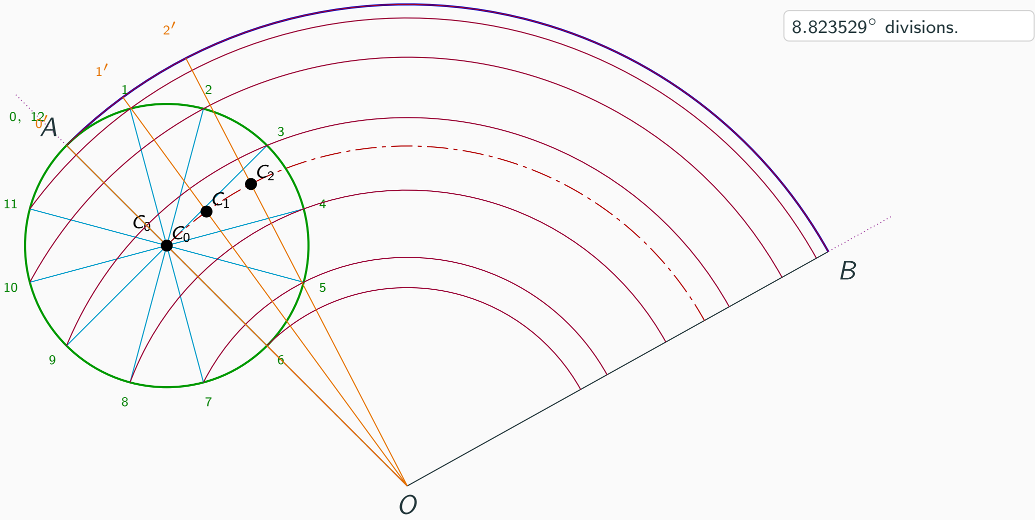
8.823529° divisions.
A
B
O
C_0
C_1
C_2
0, 12
1
2
3
4
5
6
7
8
9
10
11
0′
1′
2′

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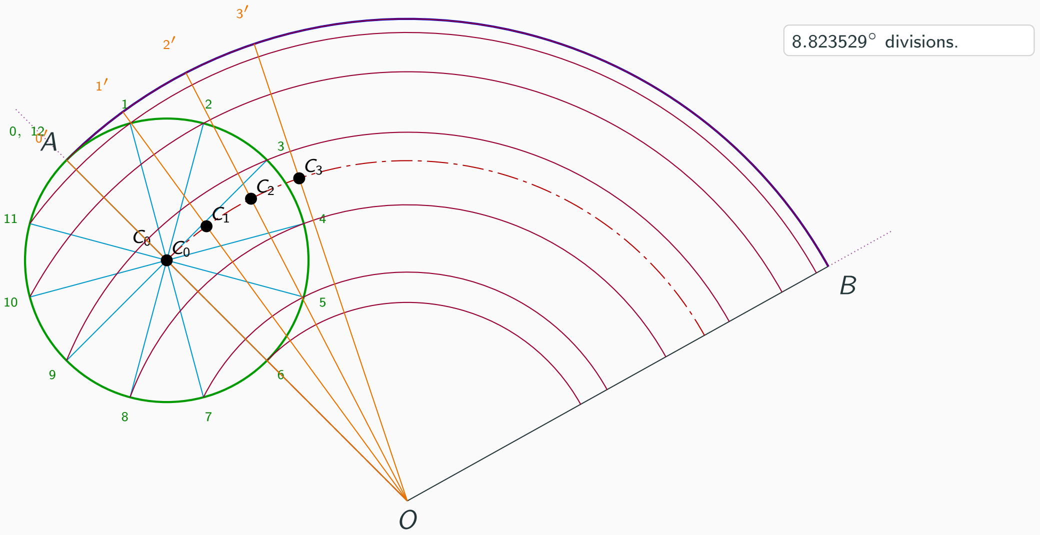

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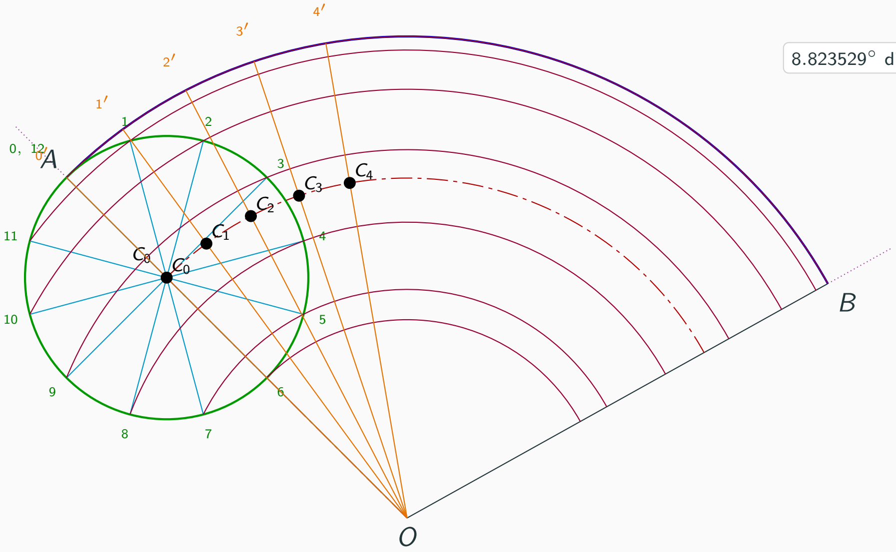

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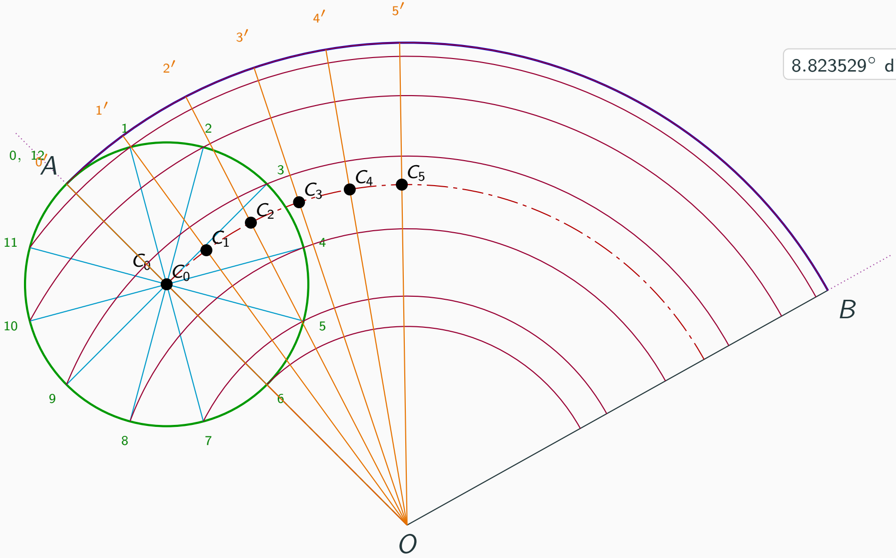

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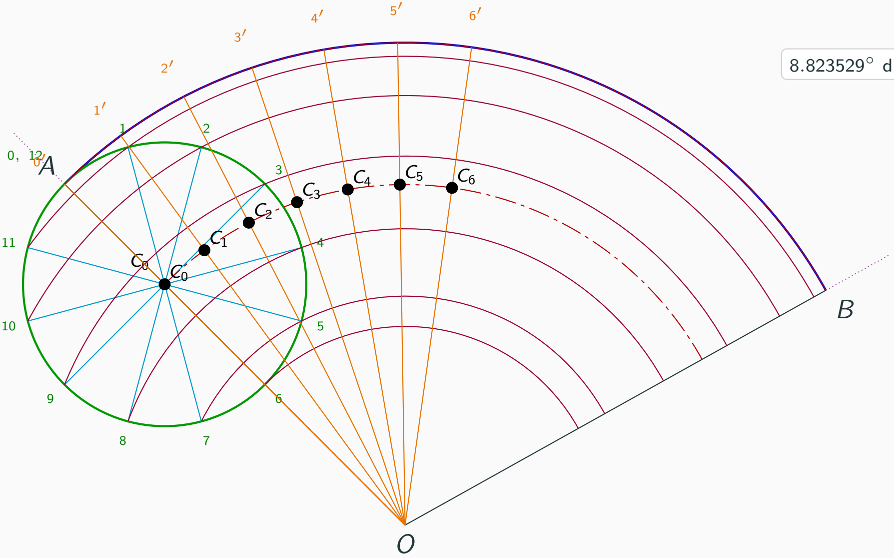

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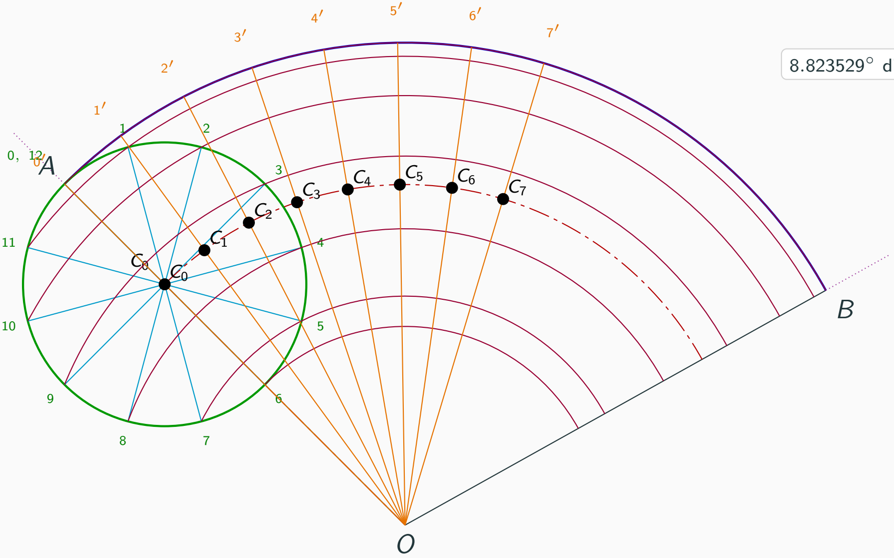

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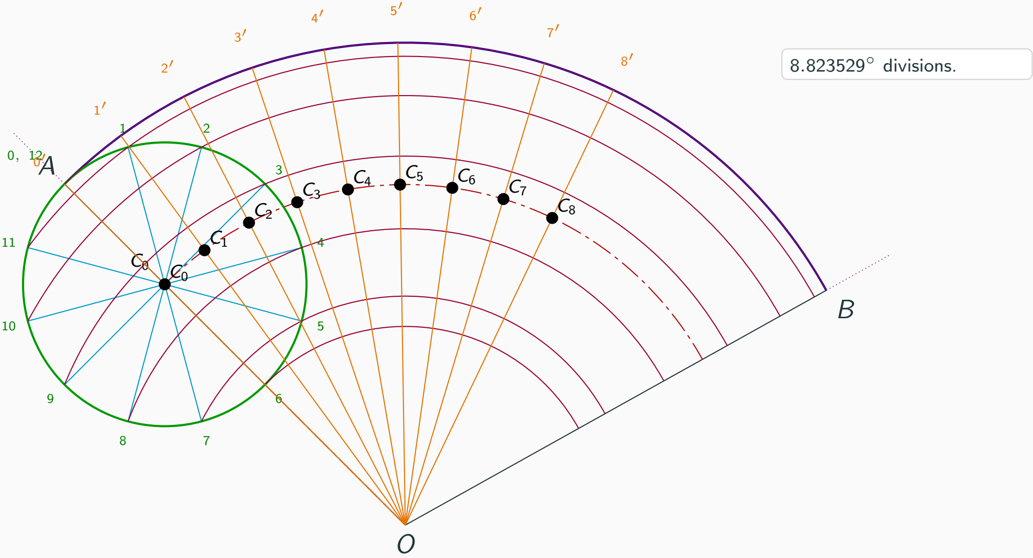
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C_7
C_8
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1
2
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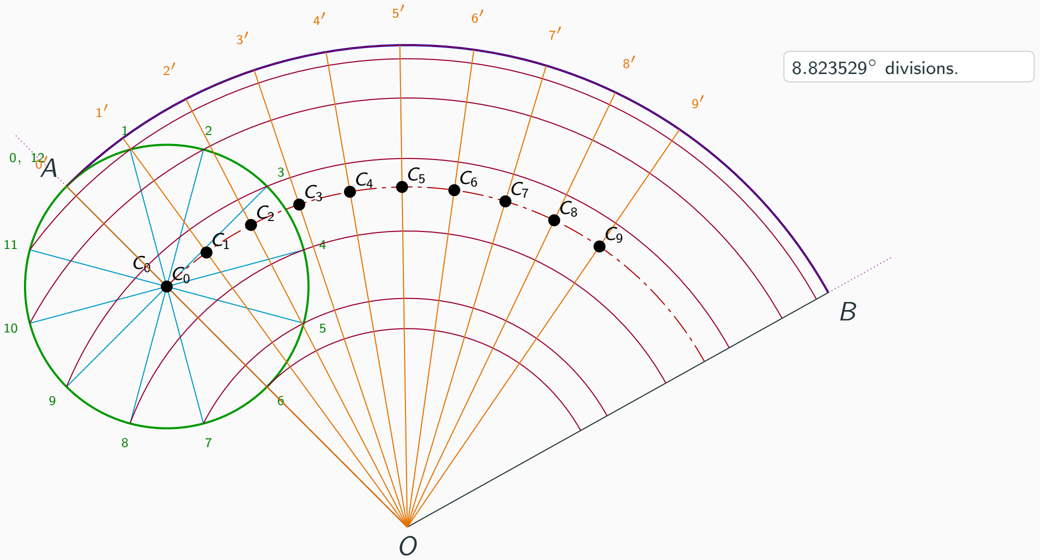
11
0′
1′
2′
3′
4′
5′
6′
7′
8′



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A
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0, 12
1
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0′
1′
2′
3′
4′
5′
6′
7′
8′
9′
C_0
C_0
C_1
C_2
C_3
C_4
C_5
C_6
C_7
C_8
C_9

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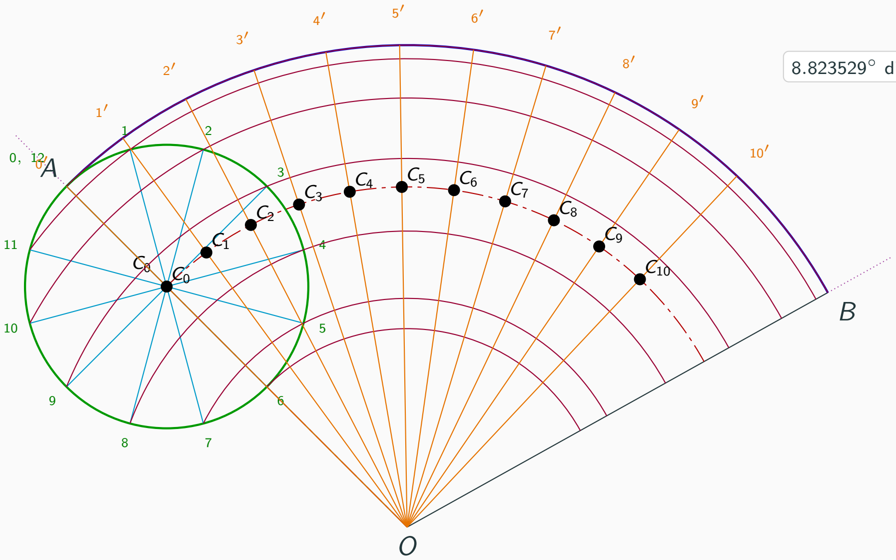

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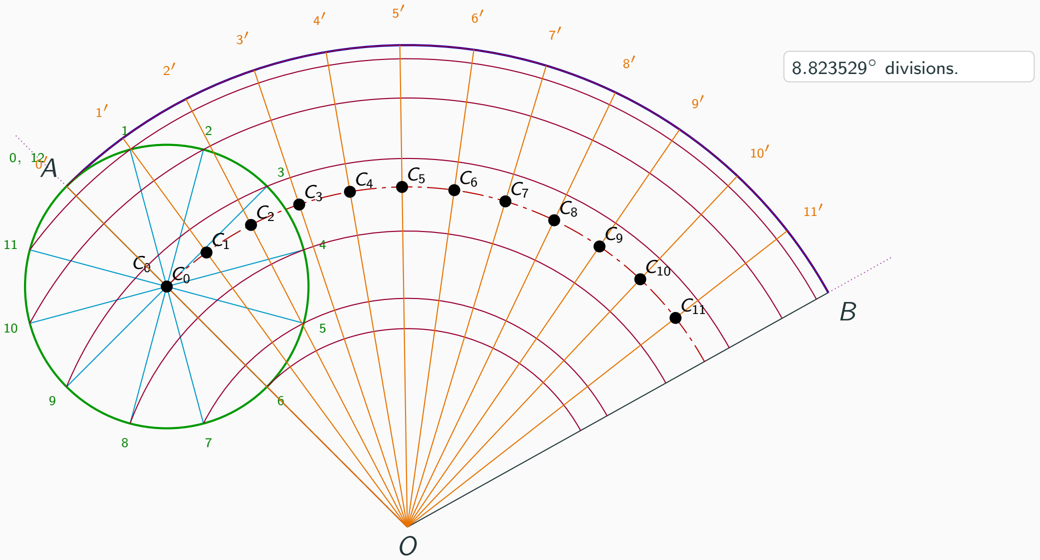

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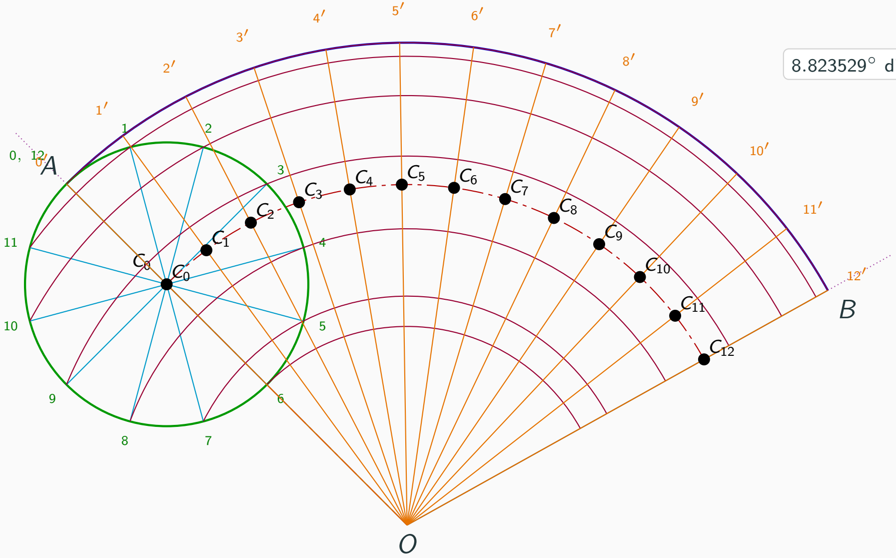
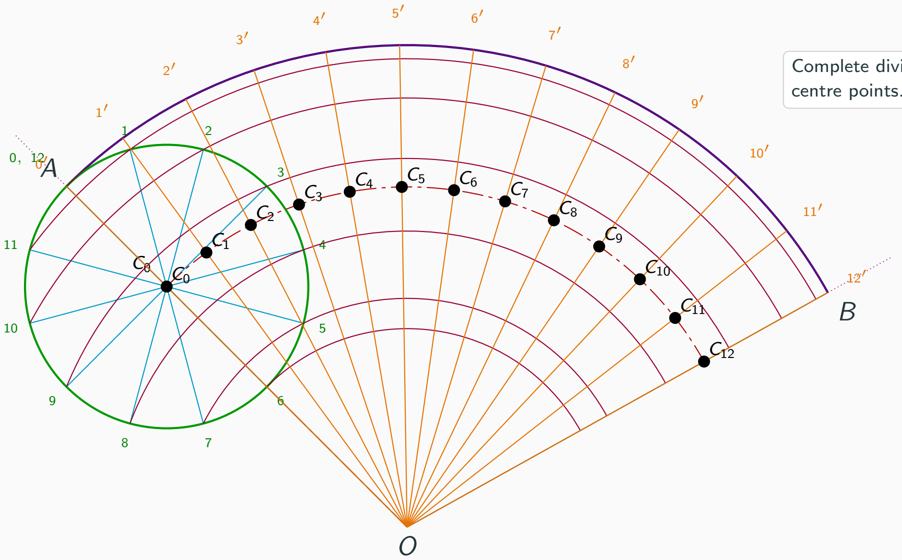


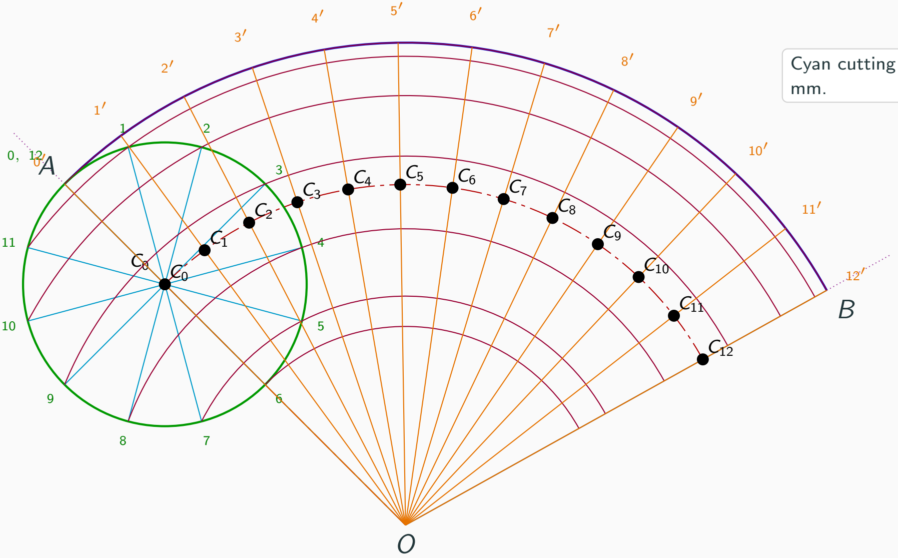
8.823529° divisions.

Step 10: Complete the angular divisions and centre points



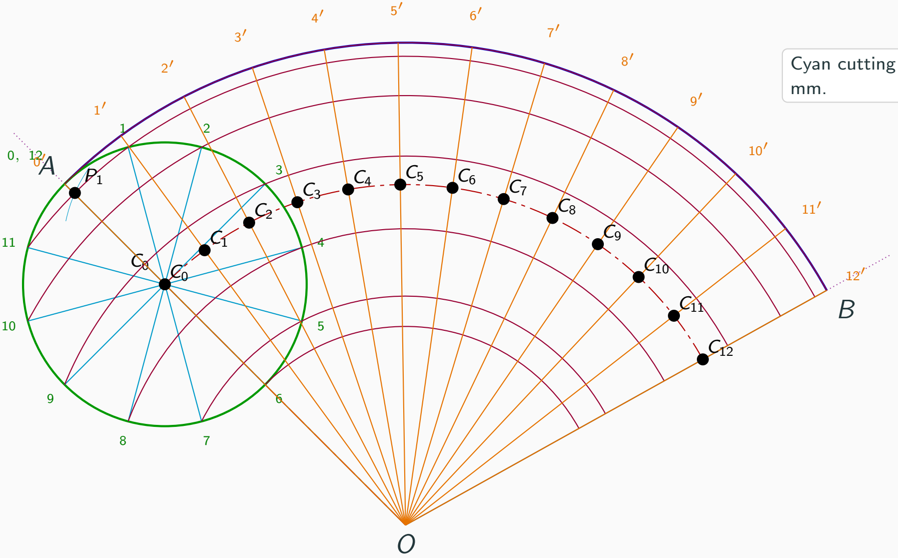
Complete division lines and centre points.

Step 11: Draw the cutting arc and locate P_i



Cyan cutting arc: radius 25 mm.

Step 11: Draw the cutting arc and locate P_i



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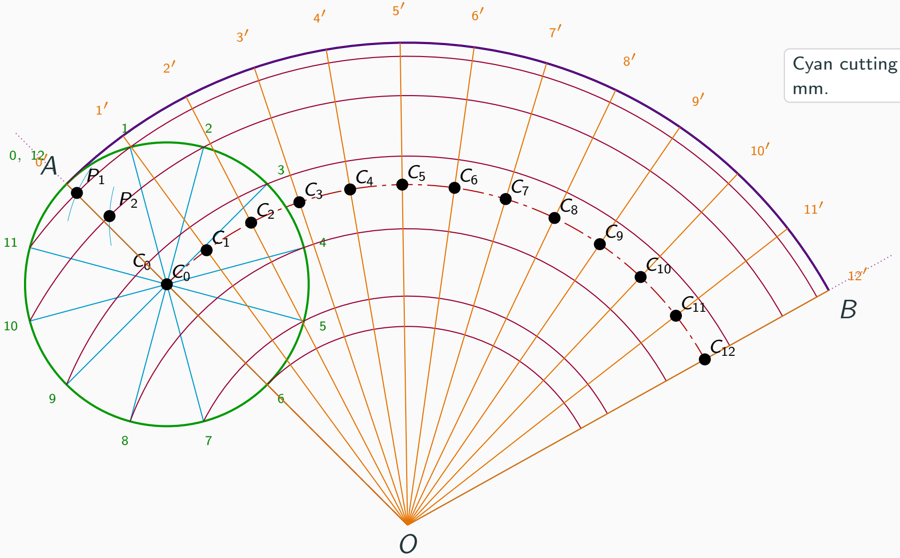

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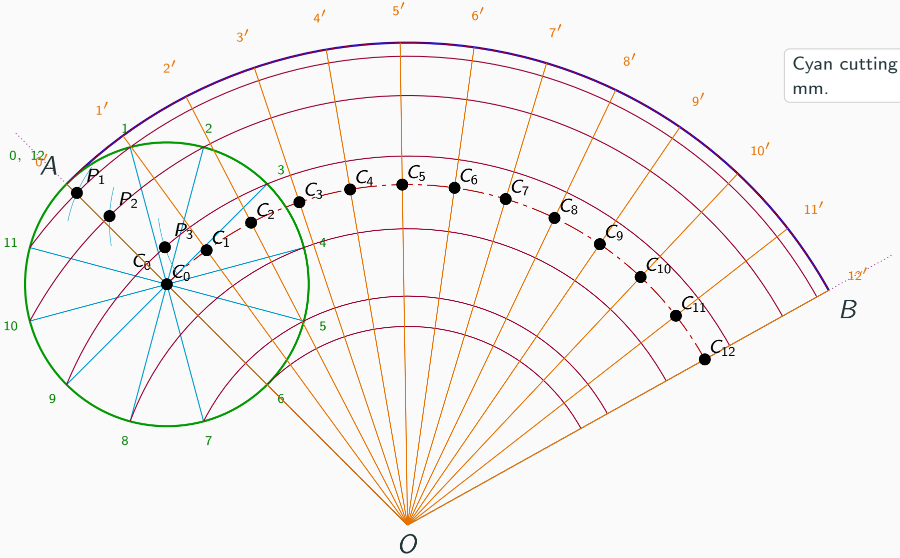

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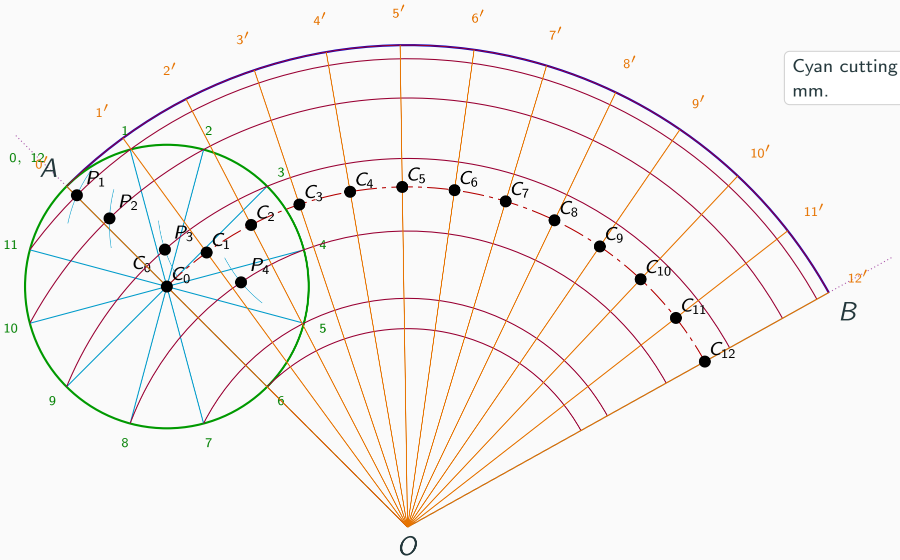

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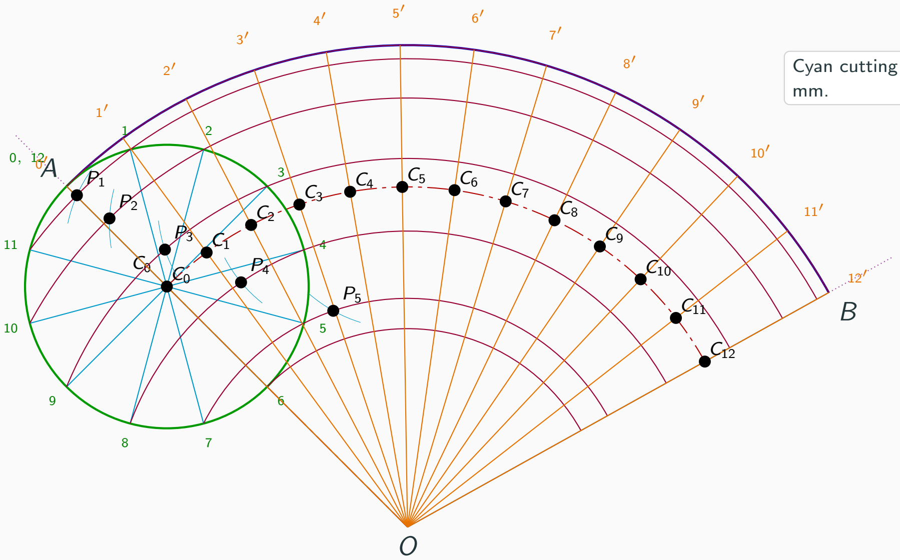

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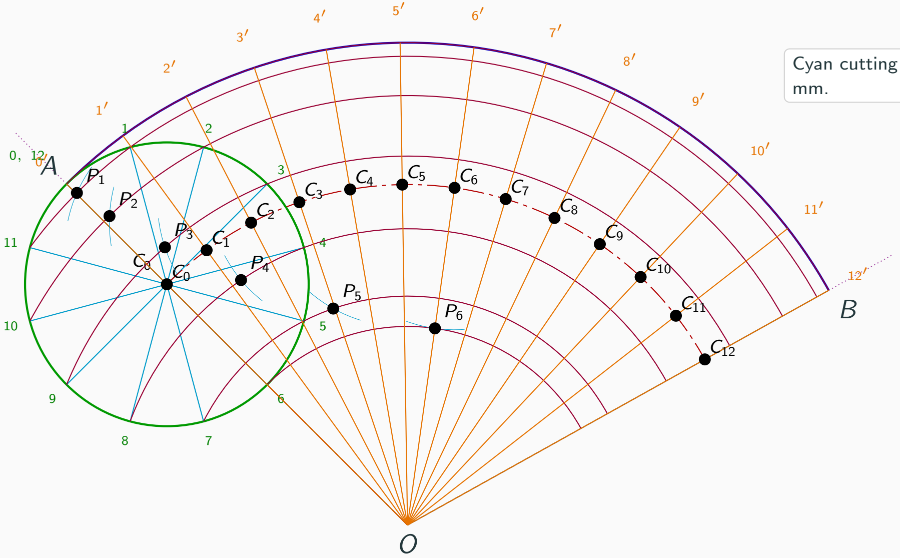

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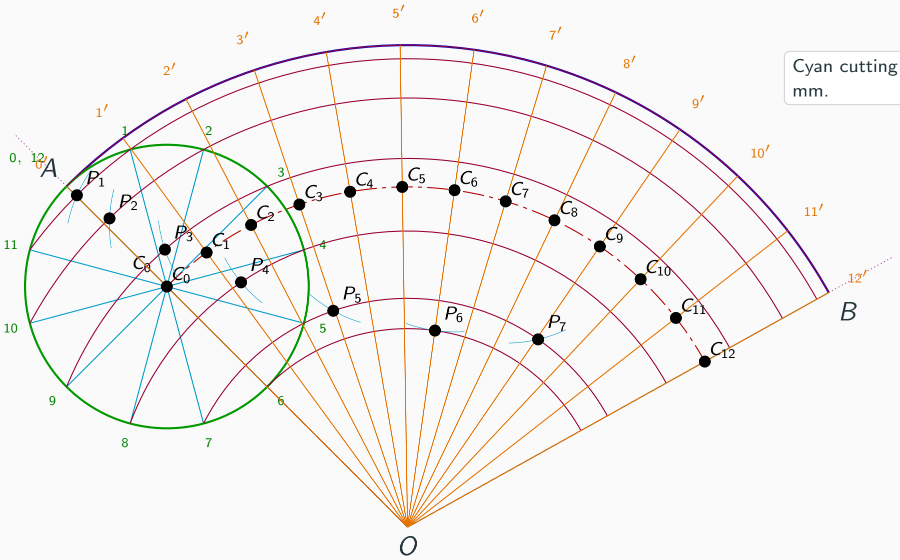

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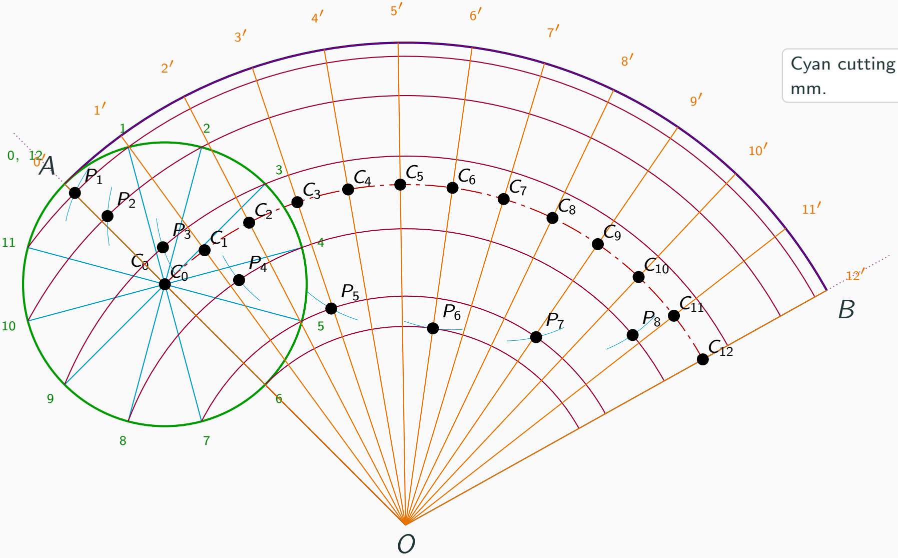

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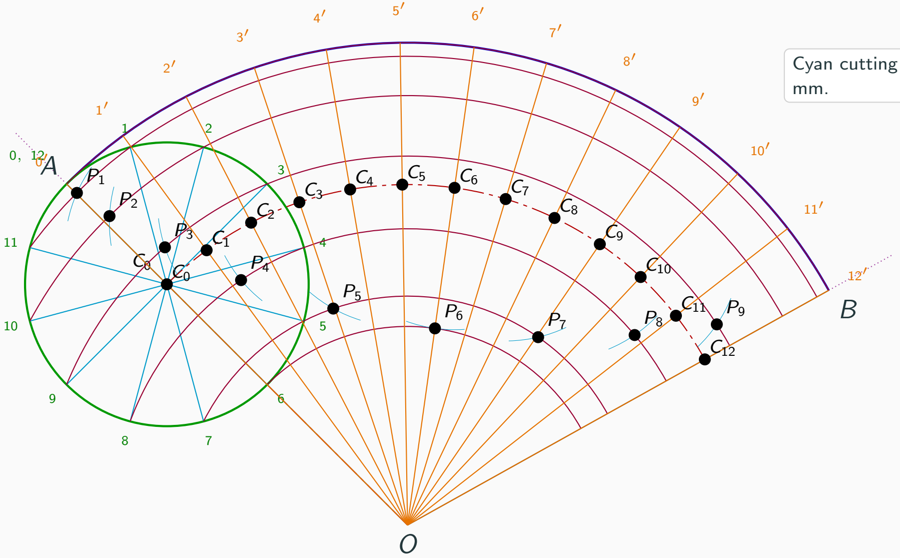

Cyan cutting arc: radius 25 mm.

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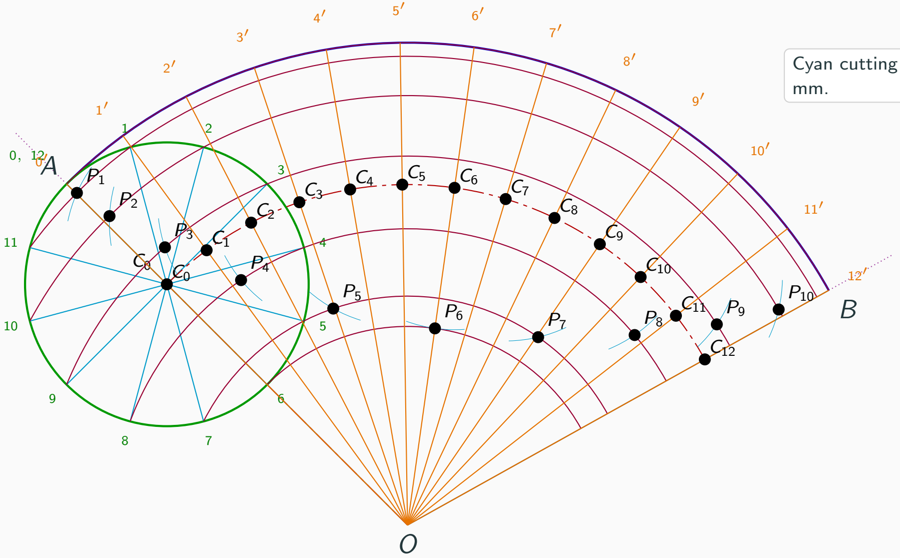

Cyan cutting arc: radius 25 mm.

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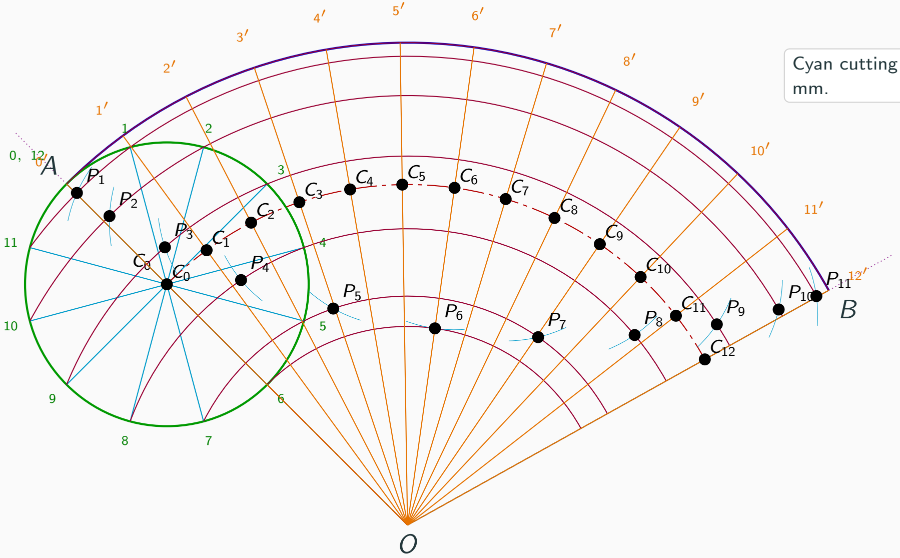

Cyan cutting arc: radius 25 mm.

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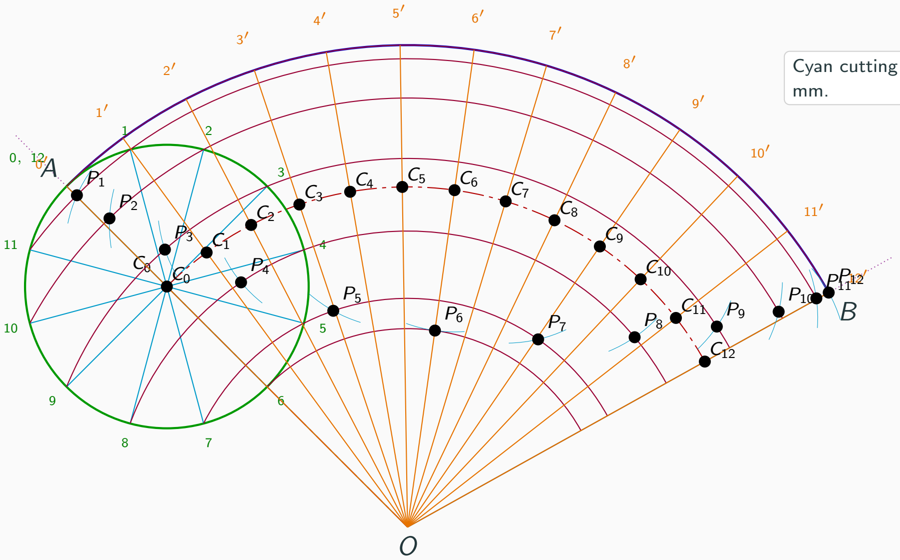
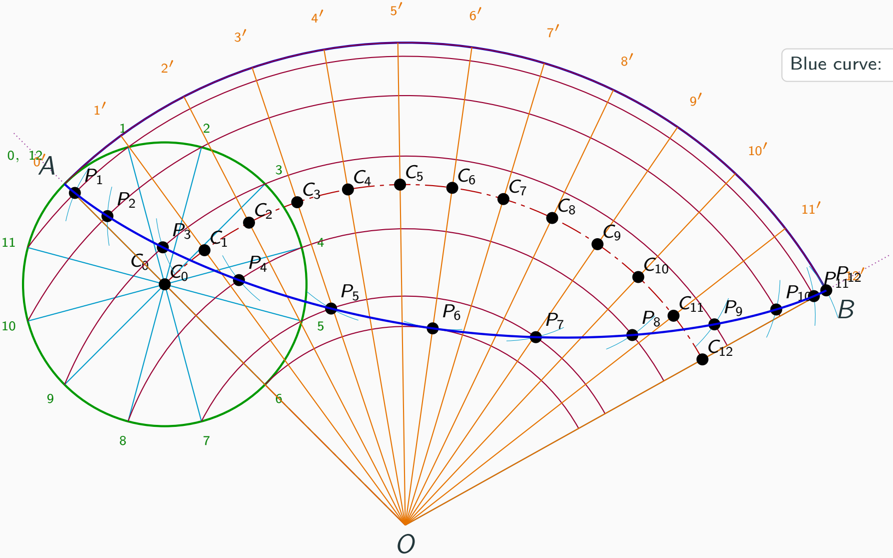


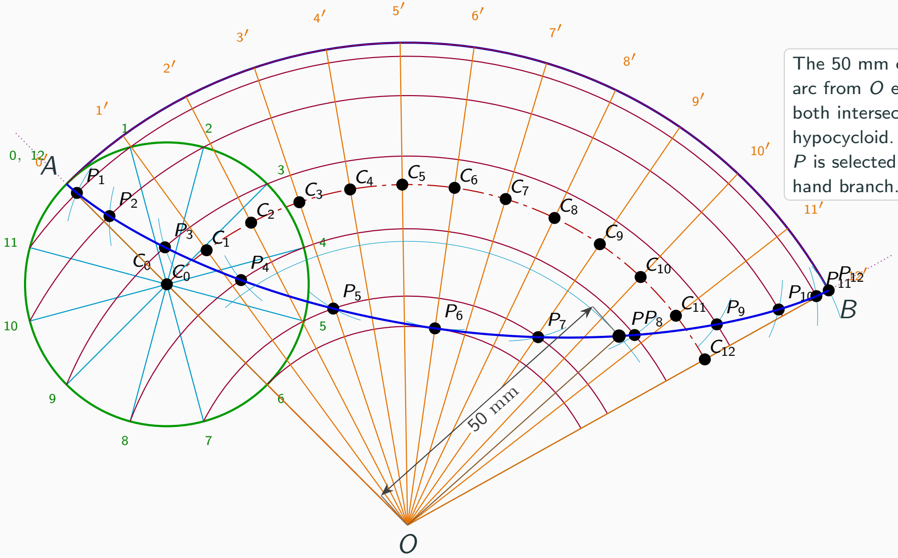
Cyan cutting arc: radius 25 mm.

Step 12: Draw the hypocycloid



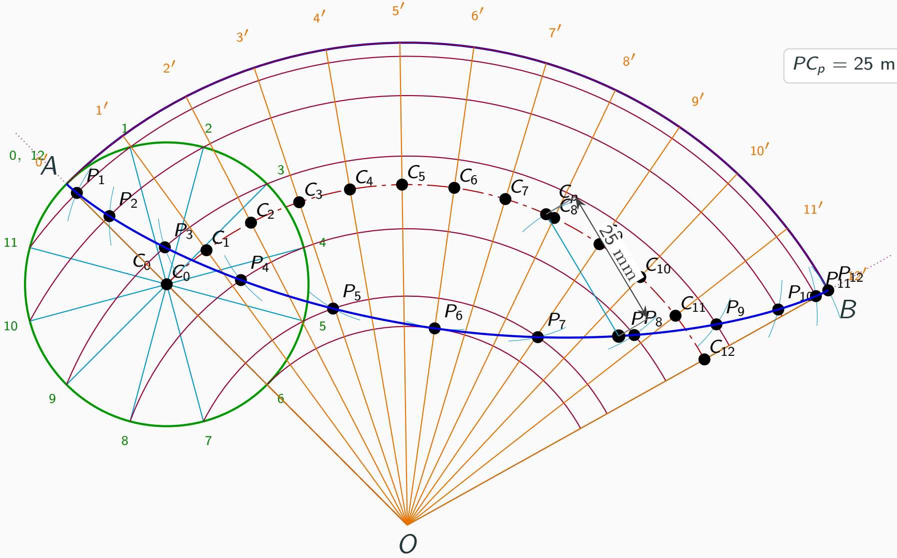
Blue curve: hypocycloid.

Step 13: Locate P at 50 mm from O



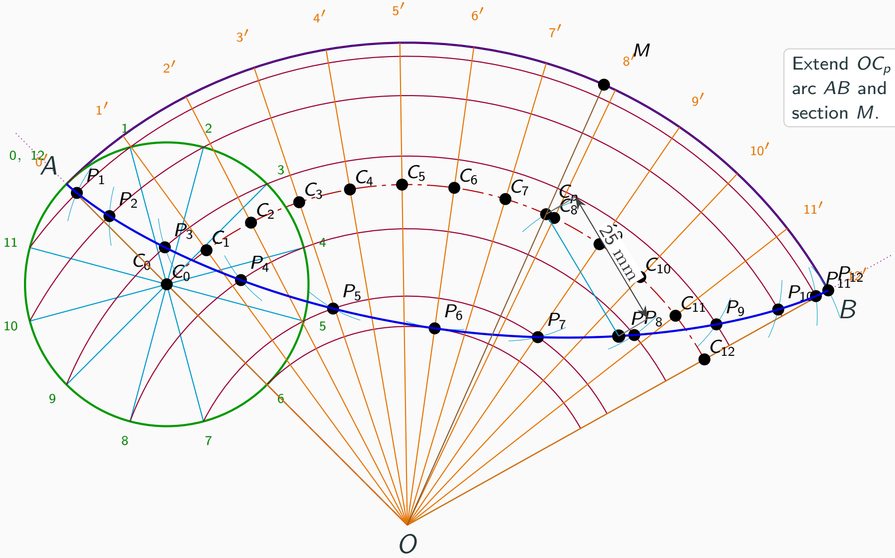
The 50 mm construction arc from O extends beyond both intersections with the hypocycloid.
P is selected on the right-hand branch.

Step 14: Draw the 25 mm radius to C_p



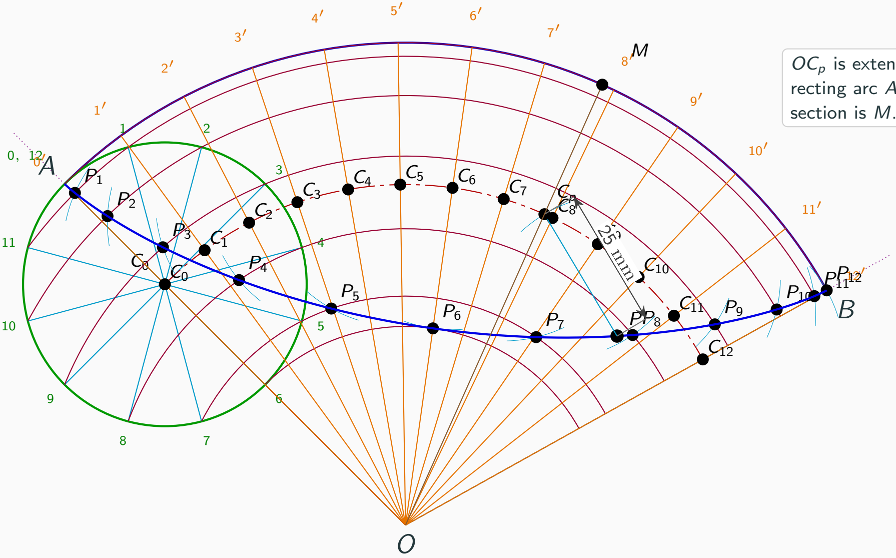
$PC_p = 25$ mm.

Step 15: Extend OC_p to the directing arc AB and mark the intersection M



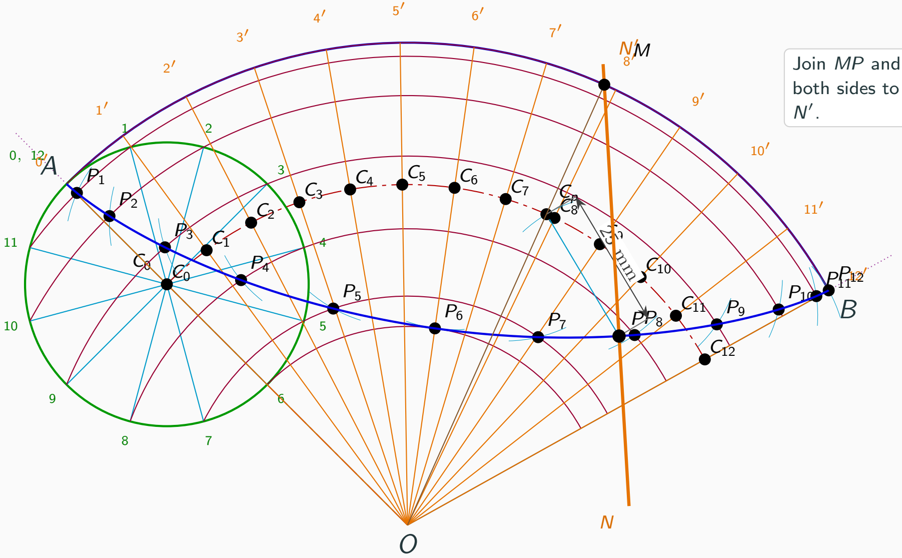
Extend OC_p to the directing arc AB and mark the intersection M.

Step 16: Mark M on the directing circle



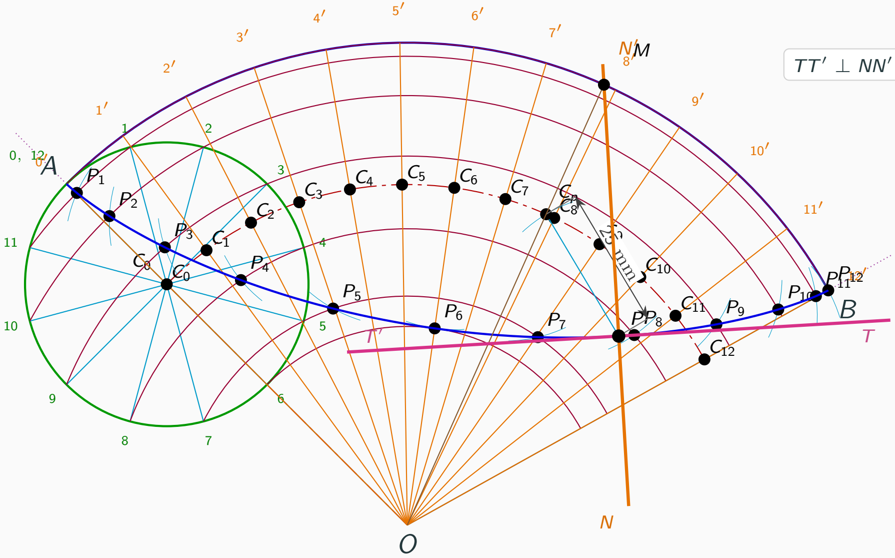
OC_p is extended to the directing arc AB; the intersection is M.

Step 17: Draw the normal NN' through P

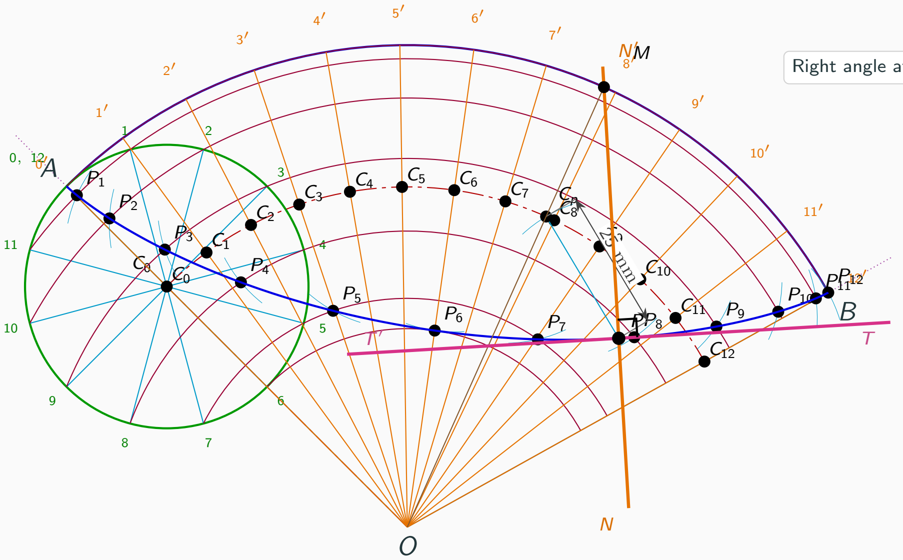


Join MP and extend it on both sides to obtain N and N'.

Step 18: Draw the tangent TT' perpendicular to NN'

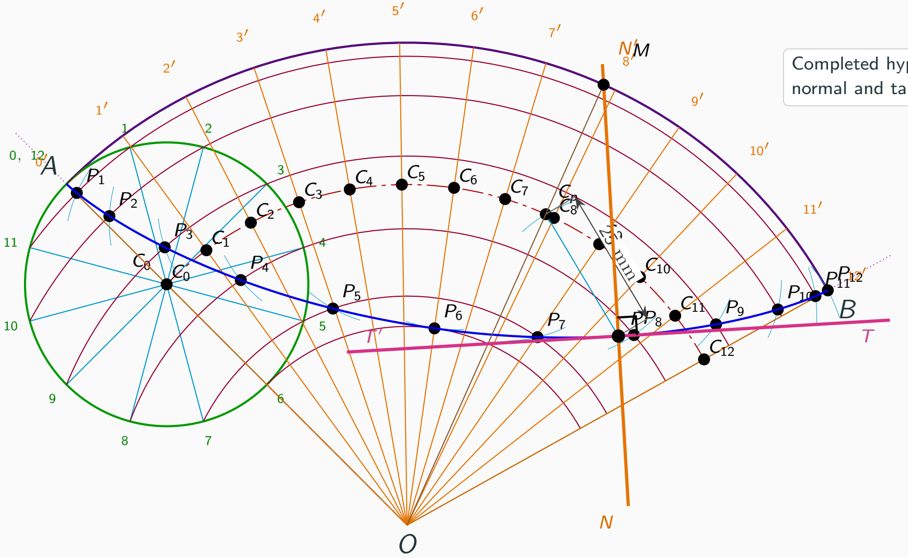


$TT' \perp NN'$ at P.



Right angle at P.

Step 20: Final completed hypocycloid construction



Completed hypocycloid with normal and tangent.